

Introduction

Biotechnology, as you would have learnt from the previous chapter, essentially deals with industrial scale production of biopharmaceuticals and biologicals using genetically modified microbes, fungi, plants and animals. The applications of biotechnology include therapeutics, diagnostics, and genetically modified crops for agriculture, processed food, bioremediation, waste treatment, and energy production.

Three critical research areas of biotechnology are

- (i) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
- (ii) Creating optimal conditions through engineering for a catalyst to act, and
- (iii) Downstream processing technologies to purify the protein/organic compound.

Let us now learn how human beings have used biotechnology to improve the quality of human life, especially in the field of food production and health.

Application of biotechnology in agriculture

Let us take a look at the three options that can be thought for increasing food production.

- a) Agro-chemical based agriculture
- b) Organic agriculture and
- c) Genetically engineered crop-based agriculture

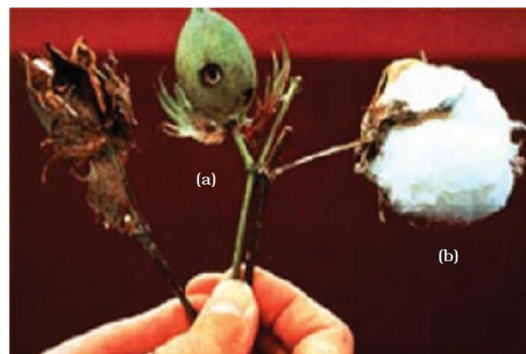
Genetically Engineered Crops

- Genetically engineered crops have desirable genes (as of insect/pest resistance, giving better yield) incorporated in them.
- Genetically modified crops have
 - More tolerance to abiotic stresses such as cold, drought, salinity, heat, etc.
 - Insect/pest resistance
 - Reduced post-harvest losses
 - Efficient mineral usage by plants
 - Enhanced nutritional value (e.g., **Golden rice**, a variety of rice is rich in Vitamin A)

NOTE – GMO (Genetically Modified Organism) – are those organisms (plants, bacteria, fungi and animals) whose genes have been altered by manipulation are called GMO)

Bt Cotton

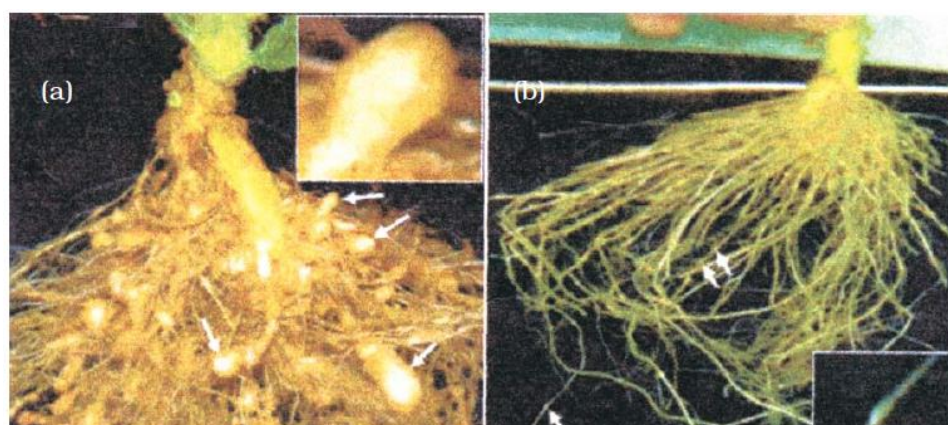
- *Bacillus thuringiensis* is a bacterium that produces proteins to kill certain insects such as lepidopterans (armyworm), coleopterans (beetles), and dipterans (flies/ mosquitoes). *B. thuringiensis* produces a protein crystal containing a toxic protein (insecticidal protein) (inactivated state - protoxin).
- Inactivated toxin (protoxin) $\xrightarrow{\text{alkaline pH}}$ Activated toxin (gut of insect)
- Activated toxin binds to the epithelial cells in the midgut of insect and creates pores that cause lyses and swelling and eventually death of insect.
- This toxin is encoded by a gene called **Cry** in the bacterium. Genes encoded by **Cry IAc** and **Cry II Ab** control cotton bollworms and those encoded by **Cry IAb** control corn borer.
- Cry genes are introduced into the cotton plants to produce Bt cotton, which is an insect resistant variety of cotton.
- Examples are Bt cotton, Bt corn, rice, tomato, potato and soyabean etc.



(a) Cotton boll destroyed by bollworms; (b) a fully mature cotton boll

RNA Interference (RNAi)

- RNAi is a method adopted to prevent infestation of roots of tobacco plants by a nematode *Meloidegryne incognitia*.
- In RNAi, a complementary RNA binds to mRNA to form a ds RNA, which cannot translate and hence, its expression is blocked (Silencing).
- This complementary mRNA may come from
 - infection by RNA viruses
 - transposons (mobile genetic elements)
- RNAi exists naturally in eukaryotes as a method of cellular defence.
- Using *Agrobacterium* vectors, nematode specific genes (DNA) were introduced in the host plant.
- The introduced DNA forms both sense and anti-sense RNA.
- Two strands being complementary to each other bend and form ds RNA, leading to RNAi.
- mRNA of nematode is silenced and the parasite cannot survive in the transgenic host.



Host plant-generated dsRNA triggers protection against nematode infestation: (a) Roots of atypical control plants; (b) transgenic plant roots 5 days after deliberate infection of nematode but protected through novel mechanism.

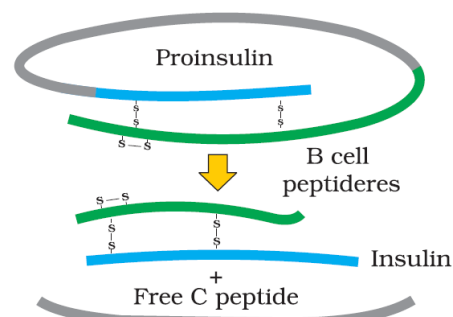
Application of biotechnology in medicine

Recombinant Therapeutics

- With the help of RDT, mass production of efficient therapeutic drugs can be accomplished.
- These are safe and do not induce unwanted immunological response as in common in case of similar products isolated from non-human sources.
- At present, about 30 recombinant therapeutics have been approved for human use the world over.
- In India, 12 of these are presently being marketed.

Genetically Engineered Insulin

- Insulin is in great demand due to increase in number of patients with adult onset diabetes.
- Insulin extracted from animal source (example, slaughtered cattle and pigs) induce allergy in humans.
- Insulin as a pro-hormone consists of 3 peptide chains – A, B, and C.
- Pro-hormone insulin $\xrightarrow{\text{removal C peptide}}$ Mature insulin
- Mature insulin consists of only two peptide chains – A and B. Both these chains were separately isolated and introduced in plasmids of *E. coli* to produce insulin chains.
- Separately produced chains A and B were extracted and combined by creating a disulphide bond to form mature human insulin.



Maturation of pro-insulin into insulin after removal of C- peptide (to be simplified)

NOTE – in 1983, **Eli Lilly** an American company prepared insulin and introduced them in plasmids of *E.coli* to produce insulin chains. Chains A and B were produced separately, extracted and combined by creating disulphide bonds to form human insulin.

Gene Therapy

- Gene therapy is an attempt to deal with genetic or congenital diseases.
- This aims at correction of a genetic defect by delivery of a normal gene into an individual or embryo to take over or compensate the function for a non-functional gene.
- The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. In this, the gene coding for enzyme ADA gets deleted leading to deficiency of ADA and problems in immune system.
- ADA deficiency can also be treated with:
 - Bone marrow transplantation
 - Enzyme replacement therapy
- Gene therapy for ADA deficiency:
 - Lymphocytes isolated from patient's blood are cultured in-vitro.
 - Functional ADA cDNA (using a retroviral vector) are then introduced into the cultured lymphocytes.
 - These lymphocytes are returned back to the patient's body.
- Lymphocytes are not immortal. Therefore, repeated infusion of genetically engineered lymphocytes is required.
- Permanent cure – Introduction of gene isolated from bone marrow cells producing ADA into cells at early embryonic stages

Molecular Diagnosis

- Recombinant DNA technologies, **PCR**, **ELISA** (Enzyme Linked Immuno Sorbent Assay) are some of the technologies of molecular early diagnosis.
- Early diagnosis of bacteria and virus in body, when the concentration is extremely low (at a time when the symptoms of the disease are not yet visible), can be done by PCR since it amplifies the DNA several folds.
- PCR is used to detect HIV virus in suspected AIDS patients and mutations in genes in suspected cancer patients.
- ELISA is based on antigen-antibody interactions. In the presence of an antigen, the antibody produced against it can be detected.

(OR)

- ELISA is based on the principle of antigen-antibody interaction. Infection by pathogen can be detected by the presence of antigens (proteins, glycoproteins, etc.) or by detecting the antibodies synthesised against the pathogen.
- Hybridisation with a radioactive probe – In this approach, gene is hybridized with a radioactive probe and autoradiography is used for detection. The regions where mutation is present in the gene will not appear in the photographic film since probe will not be able to bind with that part.

Transgenic Animals

- Animals that have their DNA manipulated to possess or express an extra gene are called **transgenic animals**.
- Till date, transgenic rats, rabbits, pigs, sheep, cows, and fish have been produced, although over 95% of all existing transgenic animals are mice.

Reasons for Producing Transgenic Animals

Study of normal physiology and development

- Transgenic animals serve as models to study genetics, regulation and down regulation of genes, and their corresponding effects on physiology.
- They give information about the biological role of a particular factor in the body.

Study of diseases

- They act as models to study genetic basis of diseases.
- These studies aid in finding possible treatments of diseases.
- Transgenic models exist of various human diseases such as cancer, cystic fibrosis, rheumatoid arthritis, Alzheimer's, etc.

Biological products

- Treatment of diseases often requires certain products that are expensive to make.
- Transgenic animals can be produced that have genes, coding for that particular product.
- Example – Human protein α -1-antitrypsin used to treat emphysema is isolated by this method. Similar attempts are being made for treatment of phenylketonuria (PKU) and cystic fibrosis.
- In 1997, first transgenic cow Rosie produced human protein-enriched milk (2.4 grams / litre), which contained α -lactalbumin and was nutritionally more balanced product suitable for human babies than natural cow- milk.

Vaccine safety tests

- Transgenic mice are used to test vaccines for their safety before they are used for humans.
- Example – Transgenic mice are used to check polio vaccines.

Chemical safety testing

- Transgenic animals contain genes that make them more sensitive to toxic substances than non-transgenic.
- Toxicity testing in such animals helps us to obtain results in less time.

Ethical Issues Associated with Transgenic Animals

- Indian government has set up an organization **GEAC (Genetic Engineering Approval Committee)**, which makes decisions regarding validity of GM research and its use for public utility.
- Modification which may result in the loss of biological significance of animals cannot go beyond regulation.
- Unpredictable results may be observed, if these organisms are introduced in natural ecosystem.
- Patents for transgenic varieties also create problems as many indigenous varieties are claimed by multinational companies as their own inventions.
- For example – in 1997 an American company got patent rights on new variety of Basmati rice through the US Patent and Trademark Office. This allowed the company to sell a 'new' variety of Basmati, in the US and abroad. This new variety was actually derived by Indian farmers by crossing Indian Basmati with semi-dwarf varieties.
- Similarly Neem and turmeric, which have been used for ages in Indian medicines, are also matters of dispute for patent rights.

Biopiracy

- Biopiracy is the term used to refer to the use of bio-resources by multinational companies (MNCs) and other organisations without proper authorisation from countries and people concerned without compensatory payment
- Most of the industrialized and developed nations are economically rich, but poor in biodiversity while opposite prevails for developing nations. Therefore, developed countries exploit traditional knowledge and resources of poor countries for commercialisation.
- This is a matter of injustice since inadequate compensation and benefit sharing is given to poor countries in return. Therefore, steps should be taken by developing countries to prevent this exploitation.
- The Indian parliament has recently introduced second amendment of Indian patents bill to deal with these issues.

SUMMARY

Biotechnology has given to humans several useful products by using microbes, plant, animals and their metabolic machinery. Recombinant DNA technology has made it possible to engineer microbes, plants and animals such that they have novel capabilities. Genetically Modified Organisms have been created by using methods other than natural methods to transfer one or more genes from one organism to another, generally using techniques such as recombinant DNA technology.

GM plants have been useful in increasing crop yields, reduce postharvest losses and make crops more tolerant of stresses. There are several GM crop plants with improved nutritional value of foods and reduced the reliance on chemical pesticides (pest-resistant crops).

Recombinant DNA technological processes have made immense impact in the area of healthcare by enabling mass production of safe and more effective therapeutics. Since the recombinant therapeutics are identical to human proteins, they do not induce unwanted immunological responses and are free from risk of infection as was observed in case of similar products isolated from non-human sources. Human insulin is made in bacteria yet its structure is absolutely identical to that of the natural molecule.

Transgenic animals are also used to understand how genes contribute to the development of a disease by serving as models for human diseases, such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.

Gene therapy is the insertion of genes into an individual's cells and tissues to treat diseases especially hereditary diseases. It does so by replacing a defective mutant allele with a functional one or gene targeting which involves gene amplification. Viruses that attack their hosts and introduce their genetic material into the host cell as part of their replication cycle are used as vectors to transfer healthy genes or more recently portions of genes. The current interest in the manipulation of microbes, plants, and animals has raised serious ethical questions.